

Hibiscus sabdariffa L. in the treatment of hypertension and hyperlipidemia: a comprehensive review of animal and human studies

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The effectiveness of *Hibiscus sabdariffa* L. (HS) in the treatment of risk factors associated with cardiovascular disease is assessed in this review by taking a comprehensive approach to interpreting the randomized clinical trial (RCT) results in the context of the available ethnomedical, phytochemical, pharmacological, and safety and toxicity information. HS decoctions and infusions of calyxes, and on occasion leaves, are used in at least 10 countries worldwide in the treatment of hypertension and hyperlipidemia with no reported adverse events or side effects. HS extracts have a low degree of toxicity with a LD50 ranging from 2,000 to over 5,000 mg/kg/day. There is no evidence of hepatic or renal toxicity as the result of HS extract consumption, except for possible adverse hepatic effects at high doses. There is evidence that HS acts as a diuretic, however in most cases the extract did not significantly influence electrolyte levels. Animal studies have consistently shown that consumption of HS extract reduces blood pressure in a dose dependent manner. In RCTs, the daily consumption of a tea or extract produced from HS calyxes significantly lowered systolic blood pressure (SBP) and diastolic blood pressure (DBP) in adults with pre to moderate essential hypertension and type 2 diabetes. In addition, HS tea was as effective at lowering blood pressure as the commonly used blood pressure medication Captopril, but less effective than Lisinopril. Total cholesterol, low-density lipoprotein cholesterol (LDL-C), and triglycerides were lowered in the majority of normolipidemic, hypolipidemic, and diabetic animal models, whereas high-density lipoprotein cholesterol (HDL-C) was generally not affected by the consumption of HS extract. Over half of the RCTs showed that daily consumption of HS tea or extracts had favorable influence on lipid profiles including reduced total cholesterol, LDL-C, triglycerides, as well as increased HDL-C. Anthocyanins found in abundance in HS calyxes are generally considered the phytochemicals responsible for the antihypertensive and hypocholesterolemic effects, however evidence has also been provided for the role of polyphenols and hibiscus acid. A number of potential mechanisms have been proposed to explain the hypotensive and anticholesterol effects, but the most common explanation is the antioxidant effects of the anthocyanins inhibition of LDL-C oxidation, which impedes atherosclerosis, an important cardiovascular risk factor. This comprehensive body of evidence suggests that extracts of HS are promising as a treatment of

hypertension and hyperlipidemia, however more high quality animal and human studies informed by actual therapeutic practices are needed to provide recommendations for use that have the potential for widespread public health benefit.